

# EE5239 Optimization Homework 4 Cover Sheet

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- Date assigned: Monday 10/21/2024
- Date due: **Friday 11/08/2024, mid-night.**
- This cover sheet must be signed and submitted along with the homework answers on additional sheets.
- By submitting this homework with my name affixed above,
  - I understand that late submission will not be accepted,
  - I acknowledge that I am aware of the University's policy concerning academic misconduct (appended below),
  - I attest that the work I am submitting for this homework assignment is solely my own, and
  - I understand that suspiciously similar homework submitted by multiple individuals will be reported to the Dean of Students Office for investigation.
- Academic Misconduct in any form is in violation of the University Disciplinary Regulations and will not be tolerated. This includes, but is not limited to: copying or sharing answers on tests or assignments, plagiarism, having someone else do your academic work or working with someone on homework when not permitted to do so by the instructor. Depending on the act, a student could receive an F grade on the test/assignment, F grade for the course, and could be suspended or expelled from the University.

# 1 Reading

- Reading: Textbook Section 2.1.1, 2.2, 2.3.1, 3.1 (except 3.1.2). (Second edition book)
- Reading: Textbook Section 3.1.1, 3.2, 3.3.1, 4.1 (except 3.1.2). (Third edition book)

# 2 Note

Please note that besides regular written problems and programming problems, you need to choose one more problem from either **Option 1**, which is one more theoretical problem, or **Option 2**, which is a programming problem.

# 3 Written Problems

(Total 60 points)

1. Exercise 2.1.6, 2.1.12 (a), 2.3.1, 3.1.1, in the textbook (Second edition book);
2. Exercise 3.1.6, 3.1.12 (a), 3.3.1, 4.1.1, in the textbook (Third edition book);

Note: for Exercise 2.1.6 and 2.12 (a) please read Example 2.1.2 (Optimizing over a simplex) in the book (or Example 3.1.2 for 3rd edition of the book).

3. (**Option 1**) **Divergence of the coordinate descent method.** Consider the following optimization problem

$$\begin{aligned} \min \quad f(x) := & -x_1x_2 - x_2x_3 - x_3x_1 + (x_1 - 1)_+^2 + (-x_1 - 1)_+^2 \\ & + (x_2 - 1)_+^2 + (-x_2 - 1)_+^2 + (x_3 - 1)_+^2 + (-x_3 - 1)_+^2 \end{aligned}$$

where the notation  $(z)_+^2$  means  $(\max\{0, z\})^2$ .

- Compute the following partial derivatives

$$\frac{\partial f(x)}{\partial x_1}, \frac{\partial f(x)}{\partial x_2}, \frac{\partial f(x)}{\partial x_3}. \quad (1)$$

- Verify that fixing  $(x_2, x_3)$  and optimizing over  $x_1$  yields the following solution

$$x_1 = \begin{cases} (1 + \frac{1}{2}|x_2 + x_3|)\text{sign}(x_2 + x_3), & \text{if } x_2 + x_3 \neq 0, \\ [-1, 1], & \text{otherwise} \end{cases}. \quad (2)$$

- Verify that starting from the following initial solution

$$(x_1^0, x_2^0, x_3^0) = (-1 - \epsilon, 1 + \frac{1}{2}\epsilon, -1 - \frac{1}{4}\epsilon)$$

for some  $\epsilon > 0$ , then by applying the cyclic coordinate descent algorithm with the order 1, 2, 3, 1, 2, 3, ..., the iterates will be cycling around six points  $(1, 1, -1)$ ,  $(1, -1, -1)$ ,  $(1, -1, 1)$ ,  $(-1, -1, 1)$ ,  $(-1, 1, 1)$ ,  $(-1, 1, -1)$ ,

- Verify that none of these six points is a stationary solution of the original problem.
- Based on Proposition 2.7.1 (version 2), or Proposition 3.7.1 (version 3), can you explain why the coordinate descent method diverges for the above problem?

## 4 Coding Problems

(Total 40 points)

1. This assignment is a follow-up on HW 3. We will apply the algorithms we learned so far to classify the hand writing digits. The homework is quite open ended, the objective is to give you some insights on the practical performance of different algorithms and formulations that we have learned so far. The data sets are available at the course web site.

**The problem:** Utilizing the two features that are provided by the data set, perform a few classification tasks on the data set using optimization algorithms. You can define your own tasks, for example separating digit “1” and “5”, or separating “1” and the rest.

Try to use **linear least squares** and the **logistic regression** models to do the classification. For each model, pick **one algorithm** that we have learned so far (coordinate descent, gradient descent, different stepsizes, etc). Once a model is built by a chosen model/algorithm, please plot the separating line graphically as is done in the slides. Please also test the performance of different algorithms using **test data set** (i.e., the percentages of error); see Lecture 5 slides.

**The requirement:** A report needs to be written on the work that has been performed. Important parts that need to be covered are: 1) description of the data; 2) description of the model; 3) description of the algorithm that solves the model; 4) description of the results.

**Note:** Items 2)-4) should be done for each model/algorithm pair. In item 4) both the quality of the model (in terms of the testing error) as well as the convergence speed of the algorithm (in terms of the objective reduction during training) need to be demonstrated. Also for each model, please plot the resulting separating line **for both training and testing data**, as is shown during the lecture.

Please put your codes in a zipped folder and upload to Canvas.

Also upload a separate pdf file to the Canvas.

**The data sets:** Please download from the course web site the “handwriting data.zip” folder. Please first read the “info.text” to get an idea of the overall data set. The data files that you are going to use are “feature\_test.txt” (testing data) and “feature\_train.txt” (training data). As have been explained before, here two specific features have already been computed for you (“symmetry” and “intensity”), for each of the data point.

2. **(Option 2)** Use the “**Perceptron**” method discussed in **Lecture 4** to perform the classification. Note that no matter what task you need to perform, it is usually not possible that the data sets are separable by a line. So you need to be careful when performing the Perceptron algorithm. The heuristic version that we have discussed in class needs to be used (Page 39, Lecture 4).